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Monsoon, 2024

B. Tech 1st Semester Examination

Mathematics I – Theory

Course Code: BMA23111T

Full Marks – 50

Time – 2 hours

The figure in the margin indicates full marks for the questions.

SECTION A

(Answer the following questions)

1. Choose the correct option:

(10×1 = 10)

(i) The value of $\int_{-\pi}^{\pi} \sin x \, dx$ is

- (a) π (b) 0 (c) 1 (d) $-\pi$

(ii) The value of $\Gamma(1/2)$ is

- (a) $\sqrt{\pi}$ (b) $\frac{\pi}{2}$ (c) π (d) $\frac{\sqrt{\pi}}{2}$

(iii) If $z = x \log y$, Then $\frac{\partial z}{\partial y}$ is

- (a) $\frac{x}{y}$ (b) $\frac{1}{y}$ (c) $\frac{y}{x}$ (d) $\frac{1}{x}$

(iv) The value of $\lim_{x \rightarrow 0} \frac{\cos x}{x}$ is

- (a) 0 (b) 1 (c) -1 (d) ∞

(v) If a function $f(x) = x^2$, $-1 \leq x \leq 1$, then there exist $c \in [-1, 1]$ such that $f'(c) =$

- (a) -1 (b) 1 (c) 0 (d) c^2

(vi) The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if

- (a) $p=1$ (b) $p<1$ (c) $p>1$ (d) none of these

(vii) A square matrix to possess the inverse if the matrix is

- (a) singular (b) diagonal (c) triangular (d) non-singular

(viii) The rank of a matrix in which every element is unity is

- (a) 2 (b) 1 (c) 3 (d) none of these

(ix) A vector $\vec{f}$ is solenoidal if

- (a) $\text{Curl } \vec{f} = 0$ (b) $\text{curl } \vec{f}$ is constant (c) $\text{div } \vec{f} = 0$ (d) $\text{div } \vec{f}$ is constant

(x) The value of $\beta\left(\frac{1}{2}, \frac{1}{2}\right)$ is

- (a) 1 (b) π (c) $\sqrt{\pi}$ (d) $\frac{\pi}{2}$

SECTION B

2. Answer any two:

(a) State Rolle's theorem. Verify Rolle's theorem on the following function (2+3=5)

$$f(x) = 2x^3 + x^2 - 4x - 2. \quad \text{on } [-\sqrt{2}, \sqrt{2}]$$

(b) Prove that (i) $\lim_{x \rightarrow \infty} \left(x \tan \frac{1}{x}\right) = 1$ (ii) $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} = \frac{1}{6}$ (2+3=5)

(c) Expand $f(x) = \sin x$ by applying Maclaurin's series. (5)

SECTION C

3. Answer any two:

(a) If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, examine that (i) $\text{grad } r = \frac{\vec{r}}{r}$, (ii) $\vec{\nabla} r^n = nr^{n-2} \vec{r}$ (5)

(b) If $u = \tan^{-1}\left(\frac{x^3+y^3}{x+y}\right)$, then by applying Euler's theorem show that (5)
$$x \left(\frac{\partial u}{\partial x}\right) + y \left(\frac{\partial u}{\partial y}\right) = \sin 2u.$$

(c) If $z = f(x+ay) + \phi(x-ay)$, then by applying Euler's theorem show that $\frac{\partial^2 z}{\partial y^2} = a^2 \left(\frac{\partial^2 z}{\partial x^2}\right)$. (5)

SECTION D

4. Answer any one:

(a) Examine the convergence of the series $\sum_{n=1}^{\infty} \left[\frac{1}{n^2} + \frac{1}{(n+1)^2}\right]$ (5)

(b) State limit comparison test. Examine that the series $\sum_{n=1}^{\infty} \sqrt{\frac{n}{n+1}}$ is not convergent: (2+3=5)

SECTION E

5. Answer any two:

(a) Show that $\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$ (5)

(b) Evaluate $\int_0^{\pi/2} \frac{\sin x - \cos x}{\sin x + \cos x} dx$ (5)

(c) Applying Gamma function, evaluate $\int_0^{\infty} e^{-ax} x^{n-1} dx$ OR $\int_0^{\frac{\pi}{2}} \sin^4 x \cos^6 x dx$ (5)

SECTION F

Answer any one:

(a) Solve, $x + 2y + 3z = 6$, $2x + 4y + z = 7$, $3x + 2y + 9z = 14$ (5)

(b) Determine the rank of the matrix $\begin{bmatrix} 1 & -1 & 2 \\ 6 & -2 & 3 \\ 3 & -2 & 4 \end{bmatrix}$ (5)

(c) Find the inverse of the matrix $\begin{bmatrix} 0 & 3 & 2 \\ -1 & 4 & 2 \\ 3 & -4 & -1 \end{bmatrix}$ (5)
